

Technology Stack

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	2 Marks

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Flask Jinja templates)	Lightweight, server-rendered forms are enough for a credit-offer form.
2	Backend / Server-Side	Python, Flask	Simple routing and form handling; integrates directly with the prediction module.
3	Database / Data Storage	CSV dataset (loan_prediction.csv); pickled model (model.pkl)	Dataset size (~500k rows) doesn't need a relational database; pickled model is small.
4	ML / Data Processing	NumPy, Pandas, scikit-learn, imbalanced-learn (SMOTE), XGBoost	Standard ML ecosystem for preprocessing, class imbalance handling, and model training.
5	Visualization	Matplotlib, Seaborn	Used during EDA to understand feature distributions and relationships.
6	Cloud / Hosting / Deployment	IBM Cloud (target), local Flask dev server (current)	IBM Cloud offers straightforward hosting for the Flask app for inference.
7	Version Control	Git, GitHub	Source control and collaboration across the team.
8	Development Environment	Anaconda Navigator, PyCharm, Jupyter Notebook	Anaconda manages the Python/data-science environment; PyCharm/Jupyter for development.